

8a	greater lattice vibrations more frequent collision of electrons with lattice ions /lower drift velocity of the electrons	A	B1 B1
8bi	connect cells in series	E	B1
8bii	connect cells in parallel	E	B1
8c	Active cooling could fail/active cooling needs energy input, increasing costs or decreasing system output/ difficult to eliminate passive cooling	A	B1
8d	site panel so that there is <u>an air gap</u> around it e.g. mounts panels a small distance above the roof/in open space/clear from obstructions/ spaced out in field.	A	B1
8ei	6.40 V, 7.60 V (2 dp)	D	A1

8eii	little / no change to current at low voltages at lower temperature, greater current at higher voltage	A	B1 B1
8eiii	Best-fit straight line drawn (See graph below)	E	B1
8eiv	$V = \mu(T_R - T) + V_R$ Plotting a graph of V against T gives gradient = $-\mu$ and y- intercept = $V_R + \mu T_R$ Use of gradient to determine μ (see graph below) $\text{Gradient} = \frac{8.65 - 6.05}{6.0 - 28.0} = -0.118 \text{ (note the dp of the coordinates)}$ $\mu = 0.12 \text{ (0.118) V } ^\circ\text{C}^{-1}$ From graph, y-intercept = 9.35, i.e. $V_R + \mu T_R = 9.35$ $6.40 + 0.12 T_R = 9.35$ $T_R = 25 \text{ (24.6) } ^\circ\text{C}$	A	C1 A1 C1 A1
8ev	$(7.60 = 0.12 (25 - T) + 6.40),$ $T = 15 ^\circ\text{C}$	A	A1
8evi	Rectangle drawn below line Correct area indicated (6.0 V and 0.048 A)	A	A1
8evii	Use of area of rectangle or $P = IV$ $P_{\max} = 4.8 \times 0.10$ $= 0.48$ $= 0.5 \text{ W}$	D	B1 A1
8fi	Better chance of capturing photons/ photons of a greater range of frequencies (contained within sunlight) can be captured	A	B1
8fii	Output power increases as angle of incidence on panel decreases / The closer the angle between PV panel and incident sunlight is to 90° , the larger the output power.	A	B1

